

3- From Fitting to Generalization

Review Questions

1. Distinguish empirical risk, population risk, and the generalization gap. Why is only one of these quantities directly observable during training?
2. Why does evaluating a fitted model on the same observations that selected it tend to give an overly favorable result?
3. Give one stable and one potentially spurious cue in a cybersecurity application. How could data collection reveal the difference?
4. What is inductive bias? Through which parts of a learning system can it enter?
5. Why can the minimum training error only decrease when moving to a nested, higher-capacity hypothesis space?
6. Explain underfitting and overfitting using both training and validation performance.
7. Why does increasing k usually reduce the capacity of k NN?
8. Define statistical bias and variance at a fixed input. How do these definitions use repeated hypothetical training samples?
9. Identify the three terms in the squared-error bias-variance decomposition. Which term cannot be removed even by the optimal predictor?
10. What is the interpolation threshold in double descent? Why is the classical U-shaped curve still useful but not universal?
11. Explain how a parameter penalty can reduce effective capacity without changing the number of parameters.
12. Compare (L_1) and (L_2) regularization. Why can (L_1) produce sparse solutions?
13. Why is early stopping a form of implicit regularization? What role does patience play?
14. Why is optimal subset selection combinatorial? State one strength and one limitation of correlation filtering, forward selection, and backward elimination.
15. Distinguish the roles of training, validation, and test data. Why can validation data also be overused?
16. Explain (K) -fold cross-validation. Why must preprocessing and feature selection be repeated inside each fold?
17. Give an application for which a random split is reasonable, one requiring a grouped split, and one requiring a temporal split.
18. Construct an example showing why accuracy can be misleading under class imbalance. What different questions do precision, recall, and (F_1) answer?
19. What does it mean to fit a scalar? Why must its estimated quantities come only from training data?
20. How can learning curves distinguish underfitting, high variance, and a data-limited regime?